

Ide & Talamàs (2025)

Artificial Intelligence in the Knowledge Economy — who gains, on which condition,
and what Lean checks

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Repository 5 · Week of September 14 – 19, 2026

github.com/Arriola123456/ai-05-ide

Version read: arXiv 2312.05481v11 (24 Feb 2025, 35 pp., pinned by SHA-256 in `lean/`); v12 (May 2025, 39 pp.) is the course PDF, same Propositions 1–6. Backup slides after slide 14.

1 · The paper and the agent's problem

Question. Modern AI does non-codifiable cognitive work and, unlike a human expert, can be copied across compute. Who gains when it enters an economy in which people organise into *workers* (routine problems) and *solvers* (exceptions)? The answer separates two dimensions: **capability** z_{AI} and **autonomy**.

Pre-AI economy (Section 3.1, p. 10). Knowledge $z \in [0, 1]$, density $g > 0$. A problem has difficulty $x \sim U[0, 1]$; it is solved (one unit of output) if the knowledge applied to it is $\geq x$.

- ▶ One-layer firm: an independent producer, expected output z .
- ▶ Two-layer firm: one solver s and workers $z \leq s$; a stuck worker asks the solver at time cost $h \in (0, 1)$. The solver's time is exhausted by

$$h n(z) (1 - z) = 1 \iff n(z) = \frac{1}{h(1 - z)}, \quad \text{team output } n(z) s.$$

AI (p. 11). One unit of compute (rental r) $\mapsto$ one agent of knowledge $z_{AI} \in [0, 1]$, a perfect substitute for a human with that knowledge. *Autonomous*: independent producer, worker or solver. *Non-autonomous* (Section 6): solver only. Compute is abundant relative to human time.

1 · The five firms and the equilibrium (p. 12–13)

Taking the wage schedule $w(\cdot)$ and r as given, a firm chooses the most profitable structure:

$$\begin{aligned}\Pi_1 &= z - w(z) \quad \text{or} \quad z_{AI} - r, & \Pi_2^{tA}(z) &= n(z)[z_{AI} - w(z)] - r, \\ \Pi_2^{bA}(s) &= n(z_{AI})[s - r] - w(s), & \Pi_2^{nA}(s, z) &= n(z)[s - w(z)] - w(s).\end{aligned}$$

Each profit is expected output minus the cost of the resources used (top-automated tA : humans under an AI solver; bottom-automated bA : AI workers under a human solver; nA : humans only). No AI–AI firm: an AI solver knows what an AI worker knows.

Competitive equilibrium (Definition, p. 13). Compute allocation (μ_i, μ_w, μ_s) , occupations (I, W_p, W_a, S_p, S_a) , a matching $m: W_p \rightarrow S_p$ with solver time clearing, $\int_Y h(1-u) dG(u) = \int_{m(Y)} dG(u)$, wages w and rental r such that firms optimise at zero profit and markets clear. Efficient (First Welfare Theorem) $\Rightarrow$ wages are marginal products.

Two maintained assumptions. $h < h_0$: no independent producers before AI (p. 18). Compute abundant relative to time: some agents *must* produce independently (footnote 14) $\Rightarrow r^* = z_{AI}$ with autonomy, $r^* = 0$ without.

2 · Prices and wages with autonomous AI (Proposition 2, p. 19)

Unique, efficient equilibrium; stratification $W^* \preceq \{z_{AI}\} \preceq S^*$; positive assortative matching. AI is always an independent producer; also a **worker** if $z_{AI} \in W$ (*basic AI*), a **solver** if $z_{AI} \in S$ (*advanced AI*). Prices:

$$r^* = z_{AI}, \quad w^*(z) = z_{AI} \left(1 - \frac{1}{n(z)}\right) \quad (z \in W_a^*), \quad w^*(z) = n(z_{AI})(z - z_{AI}) \quad (z \in S_a^*), \quad w^*(z_{AI}) = z_{AI}.$$

All three come from zero profit: $z_{AI} - r = 0$; $n(z)[z_{AI} - w] - z_{AI} = 0$; $n(z_{AI})[s - z_{AI}] - w = 0$. The AI solver keeps $z_{AI}/n(z)$ of each assisted worker's output; a human supervising AI workers keeps the whole surplus $s - z_{AI}$ over $n(z_{AI})$ agents. The person with $z = z_{AI}$ *always loses*: $w^*(z_{AI}) = z_{AI} < w(z_{AI})$ because pre-AI $w(z) > z$ for all z when $h < h_0$ (Proposition 1).

2 · The main result: Propositions 5 and 6, with all their conditions

$B = \{z \leq z_{AI} : w^*(z) > w(z)\} = [0, z_b)$, $T = \{z \geq z_{AI} : w^*(z) > w(z)\} = (z_t, 1]$ (winners sit at the extremes).

Proposition 5 (p. 24, autonomous AI). There are winners at the bottom *if AI is good enough*:

$B \neq \emptyset \iff z_{AI} > \bar{z}_{AI}$, with $\bar{z}_{AI} \in \text{int } W$. There are *always* winners at the top: $T \neq \emptyset$ for all $z_{AI} \in [0, 1)$.

Proposition 6 (p. 27, non-autonomous AI). $r^* = 0$. AI is used iff $z_{AI} > w(0)$, and then only by the least knowledgeable. In any case: (1) output is strictly higher with autonomous AI; (2) non-autonomous AI creates losers; (3) near $z = 0$: $w^* \geq \max\{w, w^*\}$, strictly if $z_{AI} > w(0)$; (4) near $z = 1$: $w^* \leq w$.

Conditions, spelled out.

1. g continuous and > 0 ; $x \sim U[0, 1]$ without loss (fn. 10); at most two layers.
2. $h \in (0, 1)$ and $h < h_0$ (no independent producers pre-AI); *top always gains* also needs $z_{AI} < 1$ — a superintelligent AI makes the most knowledgeable lose (fn. 19).
3. Compute abundant relative to time (fn. 14), the same compute in both regimes.
4. An AI agent is a perfect substitute for a human of knowledge z_{AI} (autonomous case); one z_{AI} for all agents.
5. Zero profits, wages = marginal products: the equilibrium is the planner's LP (used in the numerics).

2 · Two dimensions, not one: capability decides *whether*, autonomy decides *how much*

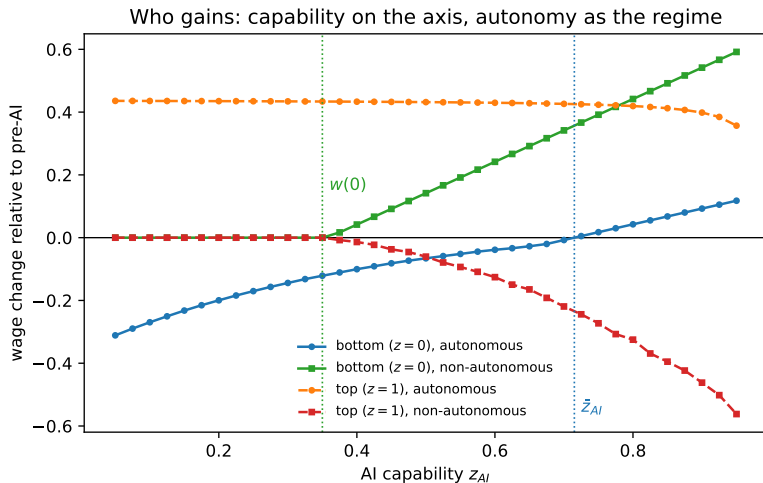
	Autonomous (co-worker + co-pilot)	Non-autonomous (co-pilot only)
Rental rate	$r^* = z_{AI}$	$r^* = 0$ (idle compute)
Bottom gains iff	$z_{AI} > \bar{z}_{AI} \in \text{int } W$ (Prop. 5)	$z_{AI} > w(0)$ (Prop. 6)
Bottom's wage when AI-assisted	$z_{AI} (1 - 1/n(z))$	z_{AI} (whole team output)
Top	always gains (Prop. 5)	may lose; $w^* \leq w^*$ at the top (6.2, 6.4)
Output	higher (Prop. 6.1)	lower

The **condition for winners at the bottom is about capability in both regimes** — a threshold on z_{AI} . What autonomy changes is the threshold ($w(0) < \bar{z}_{AI}$), the size of the bottom's gain, the fate of the top and total output. The gloss “the distributional effect is driven by autonomy, not capability” keeps the second half and drops the first. In the paper's example ($G(z) = z$, $h = 1/2$): $\bar{z}_{AI} \approx 0.715$, $w(0) = 0.358$, $\text{int } W = (0, 0.764)$.

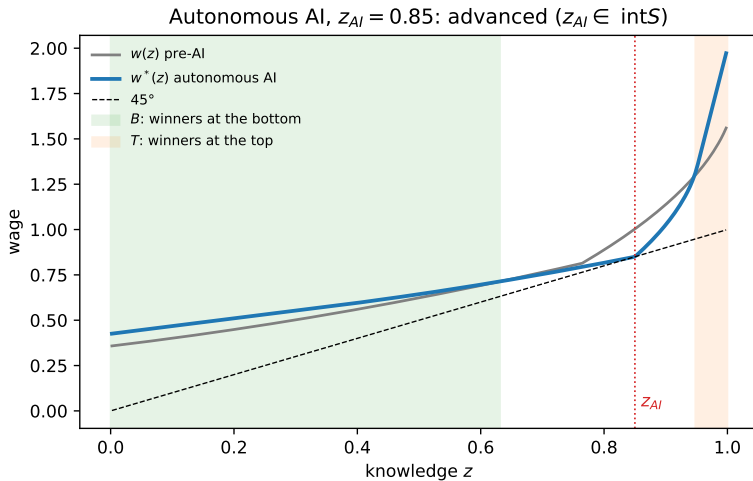
3 · What I did analytically and computationally

- ▶ **Version check.** Issue: “latest arXiv version is v11”. arXiv also has v12 (May 2025) = the course PDF (39 pp., May 20, 2025). Propositions 1–6 identical in number and statement; Lean pins v11 as instructed.
- ▶ **By hand** (hand/, backup): the firm problem, $n(z)$, the five profits, $r = z_{AI}$, the two AI wage formulas of Proposition 2, the compute accounting and the matching condition.
- ▶ **Numerics** (`analysis/knowledge_economy.py`). The equilibrium is efficient, so I solve the planner’s *linear programme* over firm configurations $(l, nA, Aind, tA, bA)$ and read $w(z)$ and r off the duals. With $G(z) = z$, $h = 1/2$, 240 types: $w(0) = 0.358$, $w(1) = 1.56$ (paper: 0.36, 1.58; closed form 1.578), $z^* = 3 - \sqrt{5}$.
- ▶ **Reproduced:** Figure 5 (no B at $z_{AI} = 0.425$; B and T at 0.85), $w^*(z_{AI}) < w(z_{AI})$, $\bar{z}_{AI} \approx 0.715 \in \text{int } W$ (Prop. 5), Prop. 6 item by item, and the AI roles of Prop. 2 (0.425: worker + independent; 0.85: all three).
- ▶ **Two- and three-type discrete versions:** closed forms (README) checked against the LP; the two-type thresholds $z_L/(1 - h(1 - z_L))$ vs z_L make the two dimensions explicit.
- ▶ **Lean via AppliedModelingLib** (slides 9–11): ten source-facing statements proved; status partially formalized.

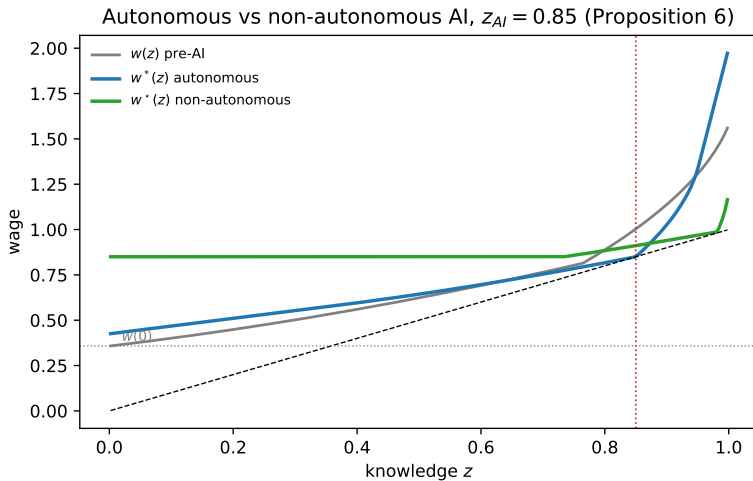
3 · Who gains: capability on the axis, autonomy as the regime



3 · Advanced autonomous AI ($z_{AI} = 0.85$): winners at both ends (paper's Figure 5b)



3 · Autonomous vs non-autonomous, $z_{AI} = 0.85$: Proposition 6 (paper's Figure 9)



4 · Lean I: the run, the statement spec, the status

Workflow (AppliedModelingLib, clone at commit 2db7d108). Pin v11 by SHA-256 $\rightarrow$ init-spec $\rightarrow$ statement spec with 10 targets (page-level locators, literal source statements, exact Lean types) $\rightarrow$ paper_contribution.py new -statement-spec (the scaffold Lean-validates every Spec) $\rightarrow$ proofs in ProofInterface.lean, helpers and extension in MainTheorems.lean $\rightarrow$ lake build IT25KnowledgeEconomy $\rightarrow$ check IT25KnowledgeEconomy -fast. **Agent:** Claude Code, not Codex (author's decision; disclosed in lean/docs/RUN_LOG.md).

The ten Specs (lean/PaperInterface.lean), all proved, no sorry: span of control $n(z)$ as the unique solution of $hn(1 - z) = 1$ (p. 10); $n > 1$ and strictly increasing on $[0, 1)$ (p. 22); the three profit identities (p. 12); Prop. 2 prices: $r^* = z_{AI}$, $w^* = z_{AI}(1 - 1/n(z)) < z_{AI}$, $w^* = n(z_{AI})(s - z_{AI})$ (p. 19); “ z_{AI} always loses” (p. 24); non-autonomous $w^* = z_{AI}$ from $r^* = 0$ (p. 26); Prop. 6.3 comparison $w^* - w^* = h(1 - z)z_{AI} > 0$ and $w^* > w \iff z_{AI} > w$ (p. 27); Prop. 6.1 in idle-compute form (p. 27).

Status: partially formalized. Existence/uniqueness of equilibrium, the occupational partition (who is in W_a^* , S_a^* , W_a^*), the strict “ $> z$ ” in Prop. 2 and the threshold $\bar{z}_{AI}$ of Prop. 5 are *declared boundaries*; the numerics cover them. **Result:** lake build IT25KnowledgeEconomy: Build completed successfully (3302 jobs); check -fast: exit 0 (lean/docs/CHECK_FAST_OUTPUT.txt). **Blocker, stated exactly:** the renamed library's root would not rebuild on my laptop in time (12 h, disk-bound), so the paper modules import three Mathlib modules instead of import Mathlib, and the spec validation ran under those imports instead of import AppliedModelingLib; no library declaration is used (RUN_LOG.md).

4 · Lean II (required slide): Proposition 2's wage of an AI-assisted worker

Paper (p. 12, 19). Zero profit of a top-automated firm with $r^* = z_{AI}$:
 $n(z)[z_{AI} - w^*(z)] - z_{AI} = 0 \Rightarrow w^*(z) = z_{AI}(1 - \frac{1}{n(z)}), n(z) = \frac{1}{h(1-z)}.$

```
def paper_prop2_ai_assisted_worker_wageSpec : Prop :=
  ∀ h z zAI w : ℝ, 0 < h → h < 1 → 0 ≤ z → z < 1 → 0 < zAI →
    (1 / (h * (1 - z))) * (zAI - w) - zAI = 0 →
      w = zAI * (1 - 1 / (1 / (h * (1 - z)))) ∧ w < zAI

theorem paper_prop2_ai_assisted_worker_wage : paper_prop2_ai_assisted_worker_wageSpec := by
  intro h z zAI w hh hh1 hz0 hz1 hzAI hzero
  have hpos : 0 < h * (1 - z) := mul_pos hh (by linarith)
  have hn : 1 / (1 / (h * (1 - z))) = h * (1 - z) := by rw [one_div_one_div]
  have hw : w = zAI - h * (1 - z) * zAI := by
    have hh' : h ≠ 0 := hh.ne'
    have h1z : (1 - z) ≠ 0 := (sub_pos.mpr hz1).ne'
    field_simp at hzero -- clears the denominator: zAI - w = h (1 - z) zAI
    linarith [hzero]
  refine ⟨?_, ?_⟩
  · rw [hn, hw]; ring
  · rw [hw]; have : 0 < h * (1 - z) * zAI := mul_pos hpos hzAI; linarith
```

Reading. h, z, z_{AI}, w are reals; $n(z)$ is spelled out as $1 / (h * (1 - z))$; the hypothesis is the zero-profit display itself. `field_simp at hzero` is the nontrivial step: it multiplies the display by $h(1 - z)$ — it needs $h \neq 0$ and $1 - z \neq 0$ as *separate* facts — then `linarith` isolates w . Lean verifies the formula *exactly* from zero profit and $r^* = z_{AI}$, and adds $w^* < z_{AI}$. Not verified: the paper's “ $> z$ ” on W_a^* — it needs which humans choose the AI-assisted firm, i.e. the equilibrium. `lake build; success; check -fast; exit 0.`

4 · Lean III: the two-type extension — capability thresholds in both regimes

Claim (README). Two types $z_L < z_{AI} < z_H$, L abundant. Autonomous: the bottom gains iff $z_{AI}(1 - h(1 - z_L)) > z_L \iff z_{AI} > \frac{z_L}{1 - h(1 - z_L)}$, a threshold strictly above z_L . Non-autonomous: gains iff $z_{AI} > z_L$; the non-autonomous premium is the AI solver's share $h(1 - z_L)z_{AI}$.

```
theorem two_type_bottom_gains_autonomous_iff (h zL zAI : ℝ)
  (hh : 0 < h) (hh1 : h < 1) (hL0 : 0 ≤ zL) (hL1 : zL < 1) :
  zL < zAI - h * (1 - zL) * zAI ↔ zL / (1 - h * (1 - zL)) < zAI := by
  have hden : 0 < 1 - h * (1 - zL) := by nlinarith
  rw [div_lt_iff0 hden]
  constructor <;> intro hx <;> nlinarith

theorem two_type_threshold_above_zL (h zL : ℝ) (hh : 0 < h) (hh1 : h < 1)
  (hL0 : 0 < zL) (hL1 : zL < 1) : zL < zL / (1 - h * (1 - zL)) := by
  have hden : 0 < 1 - h * (1 - zL) := by nlinarith
  rw [lt_div_iff0 hden]; nlinarith

theorem two_type_nonautonomous_bottom_premium (h zL zAI : ℝ) (hh : 0 < h) (hL1 : zL < 1)
  (hzAI : 0 < zAI) : zAI - h * (1 - zL) * zAI < zAI := by
  have : 0 < h * (1 - zL) * zAI := mul_pos (mul_pos hh (by linarith)) hzAI; linarith
```

Reading. `div_lt_iff0` turns the threshold inequality into a product inequality once the denominator $1 - h(1 - z_L) > 0$ is established (`nlinarith` from $h < 1$, $z_L < 1$). The LP gives the same threshold numerically (0.4615 for $z_L = 0.3$, $h = 1/2$). This is my extension, not a printed statement of the paper.

5 · Where I did not believe the AI

AI WAGE SCHEDULE:

$P_r \left[\text{AI solves } (x > z_{AI}) = 0 \right] \left[\begin{matrix} z_{AI} \text{ is the person's} \\ \text{Knowledge of AI} \end{matrix} \right]$

Fully Automated Firm:

- * $\Pi^{\text{AI}} = z_{AI} - f = 0$ (AI workers)
- * Given Free entry of firms $(\Pi^{\text{AI}})^+ = 0 \Rightarrow z_{AI} = f$

Human Workers + AI Solver:

- * $z_{AI} \rightarrow n(z)$ $\frac{\partial n(z)}{\partial z} = \frac{1}{z}$ $\frac{\partial n(z)}{\partial z} = \frac{1}{z}$
- * $\Pi_2^{\text{AI}}(z) = n(z)z_{AI} - n(z)w(0) - r$
- * $\Pi_2^{\text{AI}}(z) = n(z)[z_{AI} - w(0)] - r$

AI Workers + Human Solver:

- * $\frac{1}{n(z)} \rightarrow n(z)$ $\frac{\partial n(z)}{\partial z} = \frac{1}{z}$ $\frac{\partial n(z)}{\partial z} = \frac{1}{z}$
- * $\Pi_2^{\text{AI}}(z) = \frac{1}{n(z)}[z_{AI} - w(0)] - r$
- * $\Pi_2^{\text{AI}}(z) = \frac{1}{n(z)}[z_{AI} - w(0)] - r$

THE PROBLEM:

The firm has to choose between:

Claim 1 (ChatGPT, closing summary of the study session, prompts.md Session 1): “AI capability determina la dirección de la reasignación ocupacional; *AI autonomy determina quién captura principalmente las ganancias.*” — the one-dimensional gloss. Against Proposition 5: with autonomous AI the bottom captures gains *only if* $z_{AI} > \bar{z}_{AI}$; against Proposition 6: without autonomy, only if $z_{AI} > w(0)$. **Verdict: incomplete** — autonomy sets who gains *more*; capability sets whether the bottom gains at all.

Claim 2 (ChatGPT): $w^*(z) = z_{AI}(1 - 1/n(z))$ and $w^*(s) = n(z_{AI})(s - z_{AI})$ “from zero profit”. Checked by hand (photo) and in Lean: **correct**, and the same algebra gives $w^*(z) = z_{AI}$ when $r^* = 0$.

Claim 3 (my own LP). LP duals = equilibrium wages. Against Proposition 1’s closed form ($G(z) = z$, $h = 1/2$): $w(1) = 1.578$ vs LP 1.56 (grid ends at 0.998), $w(0) = 0.357$ vs 0.358. **Verdict: right.**

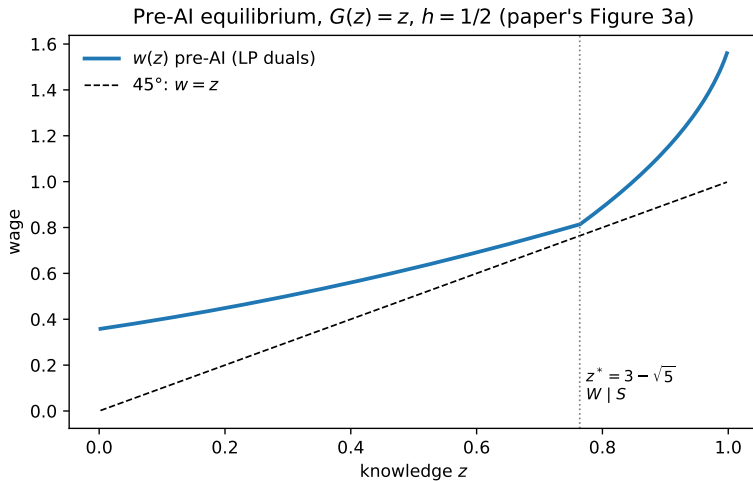
Claim 4 (Lean pushed back). Proposition 2’s “ $w^*(z) = z_{AI}(1 - 1/n(z)) > z$ ”: from zero profit alone the “ $> z$ ” is *not* provable (it depends on who is in W_a^*). Formula proved, inequality declared a boundary.

Part 3 of the hand derivation:
 $r = z_{AI}$ and the two AI wage formulas. Full page in the backup.

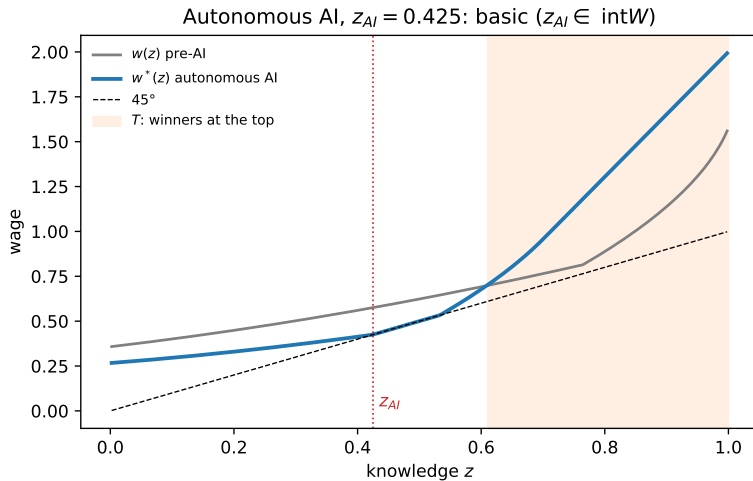
5 · The trap, the versions, and what to carry away

- ▶ **Trap.** “Driven by autonomy, not capability” collapses a 2×2 taxonomy (basic/advanced $\times$ autonomous/non-autonomous). The winners-at-the-bottom condition is a capability threshold in both regimes; autonomy moves the threshold, the size of the gain, the top’s fate and output.
- ▶ **Versions.** v11 (Feb 2025, 35 pp.) pinned for Lean; v12 (May 2025, 39 pp.) is the course PDF; JPE 133(12) 3762–3800 is the published one (ChatGPT session cites its pages). Same Propositions 1–6. Talamàs, 2025 — not the 2024 working paper.
- ▶ **Lean.** Ten Specs proved: prices and wages follow *exactly* from zero profit once r is pinned by abundant compute; the distributional propositions need the equilibrium partition, which is the declared boundary.
- ▶ **Numerics.** The efficient equilibrium is an LP; its duals reproduce the paper’s figures and give $\bar{z}_{AI} \approx 0.715$, $w(0) = 0.358$.
- ▶ **One sentence.** A co-pilot democratises knowledge, a co-worker scales the expert’s — but only an AI good enough to beat the bottom’s outside option helps the bottom at all.

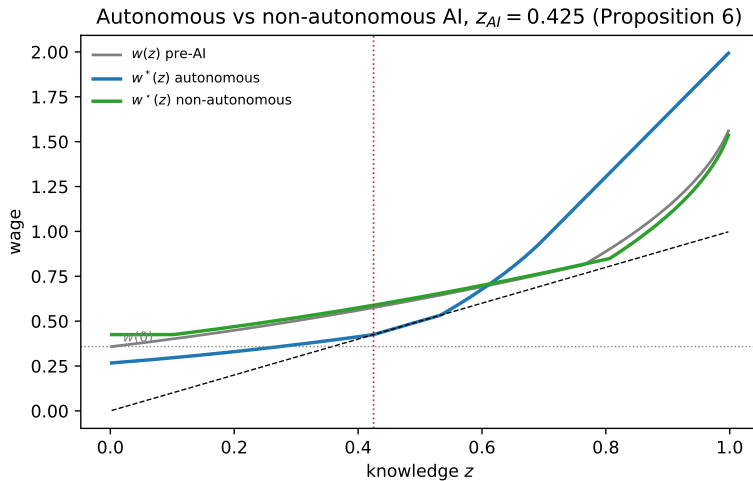
B1 · Pre-AI equilibrium, $G(z) = z$, $h = 1/2$ (paper's Figure 3a)



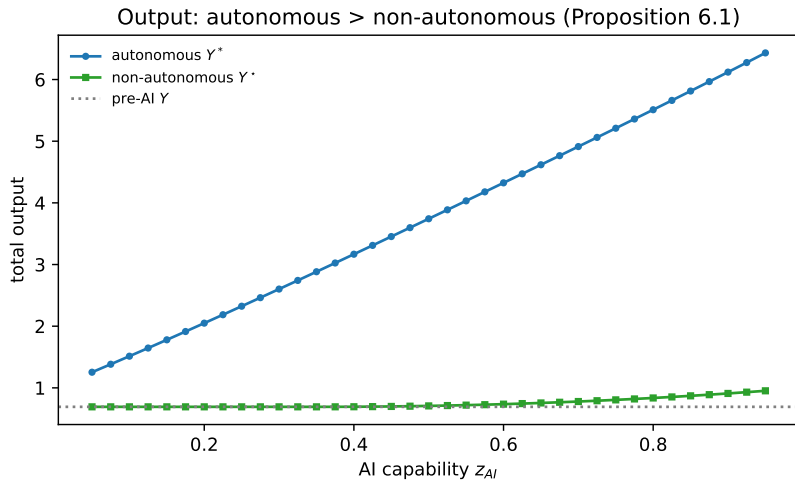
B2 · Basic autonomous AI ($z_{AI} = 0.425$): winners only at the top (Figure 5a)



B3 · Autonomous vs non-autonomous, $z_{AI} = 0.425$

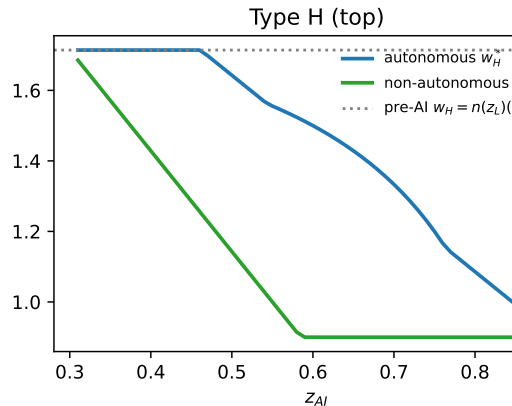
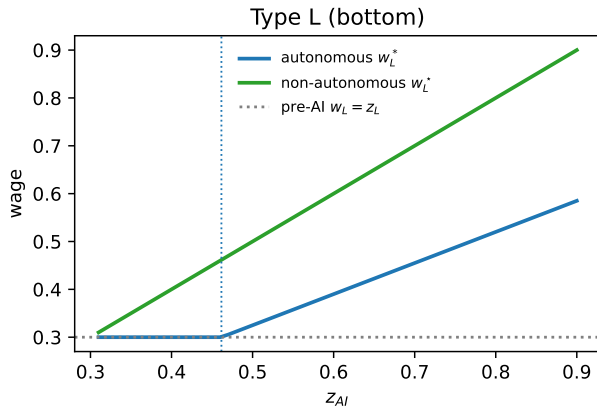


B4 · Output: autonomous above non-autonomous for every z_{AI} (Proposition 6.1)



B5 · Two-type economy: the thresholds $z_L/(1 - h(1 - z_L))$ and z_L

Two-type economy: $z_L = 0.3$, $z_H = 0.9$, $\mu_L = 0.8$, $\mu_H = 0.2$, $h = 0.5$



B6 · Hand derivation 1/5: assumptions and the pre-AI firm problem

Idé y Talamás (2025)

Supposer:

- * AI uses tacit and non codifiable knowledge
- * AI can give the firm the same AI knowledge
- * ---

Production Problem:

- * Each human has 1 unit of time and knowledge $\mathbb{E} \in [0, 1]$
- * A production opportunity has difficulty $X \sim U[0, 1]$.

* 1 output if $x \leq z$: $\Pr(x \leq z) = z$

* If $x > z$, it asks a solver with knowledge $s \geq z$:

$\Pr(\text{Ask for help}) = 1 - z$, $\Pr(\text{Success}) = s$

* One help request consumes $h \in (0, 1)$ units of solver time even if the solver fails.

* $n(z)$ identical workers:

$$\mathbb{E}[\text{Total requests}] = n(z)(1-z)$$

Total Expected time of solver: $\int n(z)(1-z) = 1$ } Normalization of solver's time (1 unit)

$$n(z) = \frac{1}{h(1-z)}$$

$\uparrow z \rightarrow \uparrow n(z)$
Each worker side knows why it fails, solver needs to consider a more worker

Firms:

$$\Pi_1^{\text{Human}}(z) = \frac{1}{z} - w(z) \quad \text{Cost}$$

$$\Pi_2^{\text{Human}}(z) = n(z)s - n(z)w(z) - w(s)$$

Production Under solver Payment to workers Payment to solver

B7 · Hand derivation 2/5: profits, competitive equilibrium, pre-AI wage schedule

$$\Pi_2^{\text{Human}}(s, z) = n(z)[s - w(z)] - w(s)$$

Competitive Equilibrium:

- * Firms choose highest Π .
- * $\uparrow \text{Firms} \Rightarrow (\Pi^* \rightarrow 0)$.
- * Workers choose highest w .
- * Capital is allocated at common price r .
- * Human and Capital clear their markets.

* Workers sort:

$$W \prec I \prec S$$

* Team output: $n(z)s$

PRE AI WAGE SCHEDULE:

Worker z and matched solver $m(z)$

$m(z) \equiv$ [The knowledge of a solver assigned to a worker of knowledge z (before this was labeled 's')]

$$\Pi_2^{\text{Human}}(s, z) = n(z)s - n(z)w(z) - w(s)$$

Free entry $\Rightarrow (\Pi_2^* \rightarrow 0)$

$$\Pi_2^{\text{Human}}(m(z), s) = n(z)m(z) - n(z)w(z) - w(m(z)) = 0$$

$$\Rightarrow m(z) = w(z) + \frac{w(m(z))}{n(z)}$$

$$w(z) = m(z) - \frac{w(m(z))}{n(z)}$$

Worker wage
Schedule
before AI

Independent workers make $w(z) = z$.

$$\frac{\partial \Pi_2^{\text{Human}}}{\partial m(z)} = n(z) - w'(m(z)) \Rightarrow \boxed{w'(m(z)) = n(z)}$$

First order condition

B8 · Hand derivation 3/5: $r = z_{AI}$ and the AI wage schedule

UNITES

Solver wage schedule

AI WAGE SCHEDULE:

$$Pr(AI \text{ solver} | x > z_{AI}) = 0 \quad \left[\begin{array}{l} z_{AI} \text{ is the general} \\ \text{knowledge of all} \\ \text{AI's} \end{array} \right]$$

Fully Automated Firm:

$$* E[output] = z_{AI}$$

$$* \pi_1^{AI} = z_{AI} - r = 0 \quad \xrightarrow{\text{cost of AI}} \quad (1/0 \text{ workers})$$

$$* \text{Given free entry of firms } (\pi_1^{AI})^* = 0 \rightarrow \boxed{z_{AI} = r}$$

Human Workers + AI Solver:

$$* IA \rightarrow n(z) \quad , \quad z \leq z_{AI}$$

Solver
Human Worker
Worker's Knowledge

$$* \pi_2^{IA}(z) = n(z)z_{AI} - n(z)w(z) - r$$

Output of (IA) human workers under AI solver
Cost of (IA) human workers
Cost of AI

$$\pi_2^{IA}(z) = n(z)[z_{AI} - w(z)] - r$$

AI Workers + Human Solver:

$$* \frac{1}{s} \text{ human} \rightarrow n(z_{AI}) \quad , \quad s \geq z_{AI}$$

Solver
AI workers
Solver's Knowledge

$$* n(z_{AI}) = \frac{1}{n(1-z_{AI})} \equiv \text{ratio of AI workers}$$

$$* \pi_2^{BAI}(s) = n(z_{AI})s - n(z_{AI})r - w(s)$$

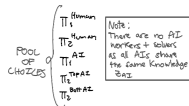
Output of (BAI) AI workers under human solver
Cost of (BAI) AI worker
Cost of human solver

$$\pi_2^{BAI}(s) = n(z_{AI})[s - r] - w(s)$$

THE PROBLEM:

The firm has to choose between:

B9 · Hand derivation 4/5: the firm's problem and the compute accounting



Also, each human must end as either:

- * I = Independent Producer
- * Wp = Human worker assisted by people
- * Wa = Human worker assisted by AI
- * Sp = Human solver assisting people
- * Sa = Human solver assisting AI

COMPUTE DISTRIBUTION:

m = Resources available for use

1 compute unit $\rightarrow$ 1 agent AI

$$\underbrace{m_I}_{\text{Compute for Independent AI}} + \underbrace{m_w}_{\text{Compute for AI Workers}} + \underbrace{m_s}_{\text{Compute for AI Solvers}} = m$$

We assume there are individuals with knowledge $z \in [0, 1]$, such that

$$z \sim \mathcal{U}(0, 1), \text{ CDF}(z) = G(z)$$

$g(z)$ is the density function.

We know that:

$$n(z) = \frac{1}{h(1-z)} \rightarrow \underbrace{\text{Use of AI per worker}}_{h(1-z)} = \frac{1}{n(z)}$$

Given this, we can define:

$$m = \int h(z) \cdot n(z) \cdot dz$$

B10 · Hand derivation 5/5: the matching condition and market clearing

$$m_s = \int_{W_A} h(1-z) \theta(z) dz$$

$$m_s = \int_{W_A} h(1-z) \underbrace{dG(z)}_{\frac{dG(z)}{dz} = g(z)}$$

Now for m_v , we know that:

$$m_v = \underbrace{n(z_{AZ})}_{\substack{\text{1 solver uses} \\ n(z_{AZ}) \\ \text{workers}}} \underbrace{\int_{S_A} dG(s)}_{\substack{\text{Aggregation} \\ \text{of all} \\ \text{human solvers}}}$$

Matching Function:

$$m: W_F \rightarrow S_F$$

$m(z)$: Knowledge of a solver assigned to worker z

Given that;

$$W_A \rightarrow \text{Has a solver with } z_{AZ}$$

$$S_A \rightarrow \text{Has workers with } z_{AZ}$$

There is only uncertainty in:

$$(z) W_F \rightarrow S_F \quad (s = m(z))$$

Equilibrium:

For y workers

$$\underbrace{\int_Y h(1-u) dG(u)}_{\text{Demand for solver time from the } y \text{ workers}} = \underbrace{\int_{m(Y)} dG(u)}_{m(Y) \equiv \text{set of solvers matched}}$$

• Each has 1 unit of time

Demand = Supply

With this we characterize equilibrium as:

$$m_A + m_W + m_S = m \quad \text{All computers are assigned}$$

$$I \cup W_F \cup W_A \cup S_F \cup S_A = [0,1] \quad \text{All workers are occupied}$$

Each Firm also chooses how to work

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def paper_span_of_controlSpec : Prop :=
  ∀ h z n : ℝ, 0 < h → h < 1 → 0 ≤ z → z < 1 →
    (h * n * (1 - z) = 1 ↔ n = 1 / (h * (1 - z)))

def paper_prop6_bottom_prefers_nonautonomousSpec : Prop :=
  ∀ h z zAI wpre : ℝ, 0 < h → h < 1 → 0 ≤ z → z < 1 → 0 < zAI →
    zAI - zAI * (1 - 1 / (1 / (h * (1 - z)))) = h * (1 - z) * zAI ∧
    zAI * (1 - 1 / (1 / (h * (1 - z)))) < zAI ∧
    (wpre < zAI ↔
      wpre < max wpre zAI ∧ zAI = max wpre zAI)

def paper_prop6_output_idle_computeSpec : Prop :=
  ∀ zAI used total Yhuman : ℝ, 0 < zAI → used < total →
    Yhuman + zAI * used < Yhuman + zAI * total

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Reading. The first Spec is the page-10 display as an equivalence: the span of control is *the* solution. The Prop. 6.3 Spec compares the three wages of a worker assisted by AI in both regimes: the non-autonomous premium over the autonomous wage is $z_{AI}/n(z) = h(1 - z)z_{AI}$ (the AI solver's share when $r^* = z_{AI}$), and the gain over the pre-AI wage is exactly the condition $z_{AI} > w$ — at $z = 0$ this is the proposition's “ $z_{AI} > w(0)$ ”. The last Spec is the accounting behind 6.1: idle compute is output forgone at z_{AI} per unit.